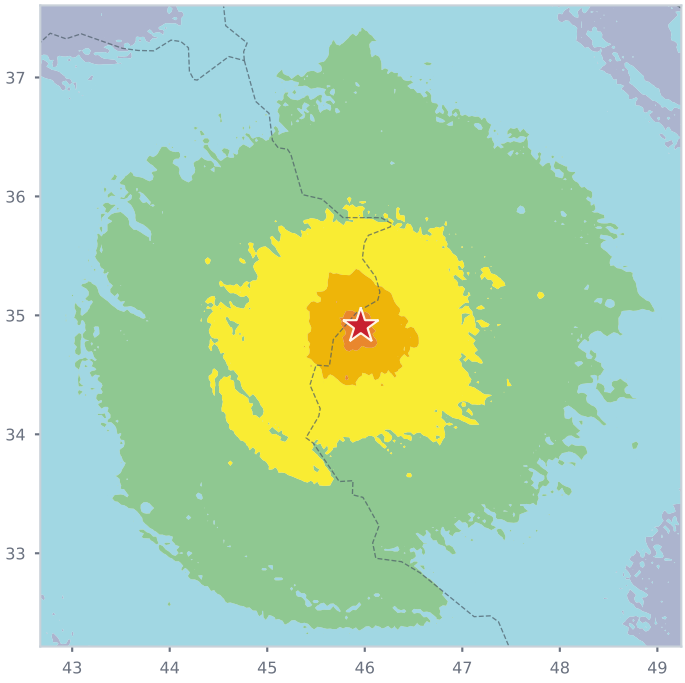
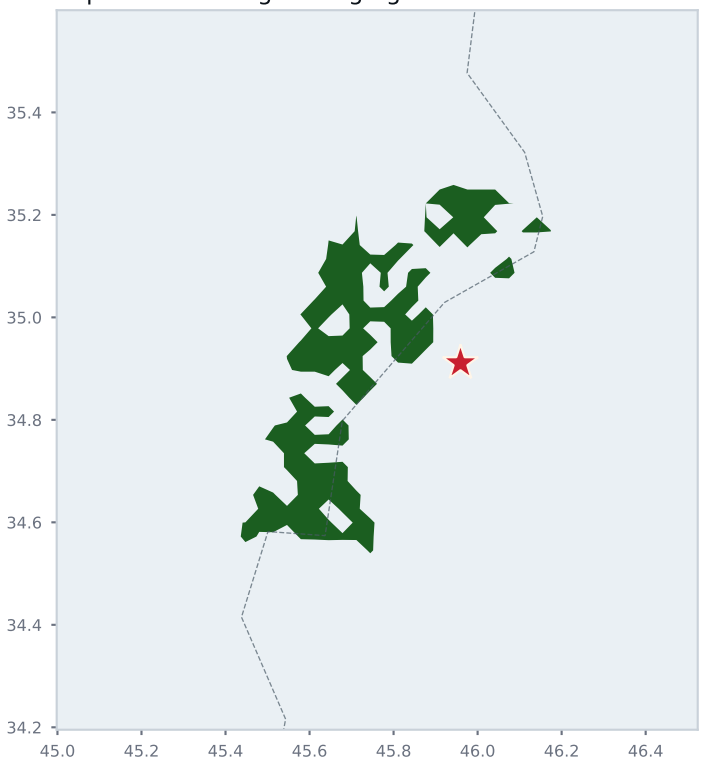
Shaking intensity (IMS-25)



Expected building damage grade



IMS-25 intensity



Expected damage grade (IMS-25)



Damage alert

RED Extensive destruction across the area

Provisional **Transit occupancy**

Buildings heavily damaged (DG3 or worse)

4.0 thousand to 83 thousand

Central estimate 18 thousand. The range is the 5th to 95th percentile.

Buildings in the shaken area

7.8 million

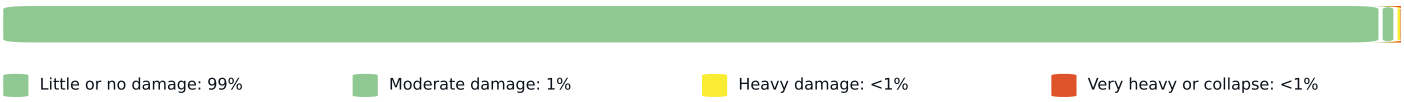
21% of them inferred from built-up area, not observed footprints.

People by shaking level

VIII	3.2 thousand
VII	326 thousand
VI	2.9 million
V	18 million
IV	11 million

Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage



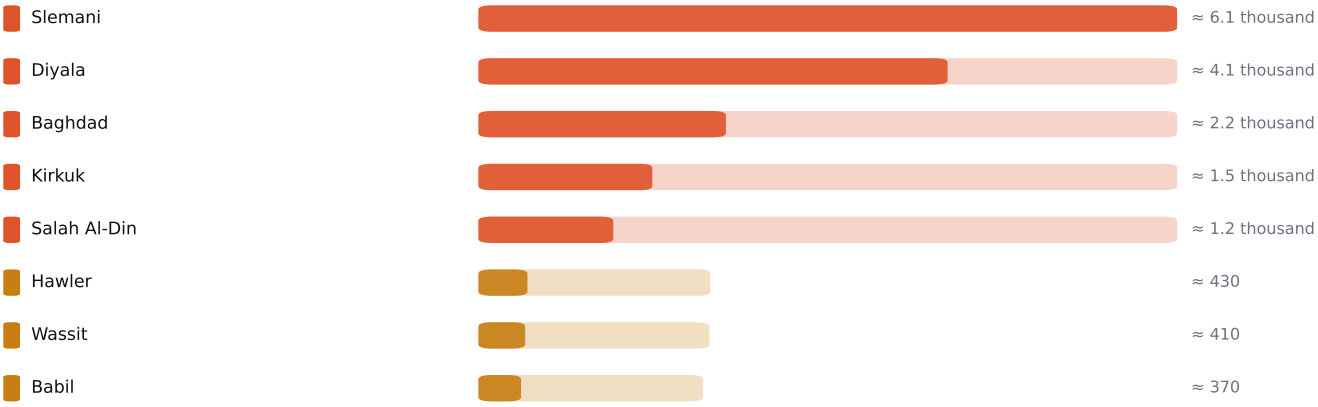
Little or no damage: 99% | Moderate damage: 1% | Heavy damage: <1% | Very heavy or collapse: <1%

Which buildings failed

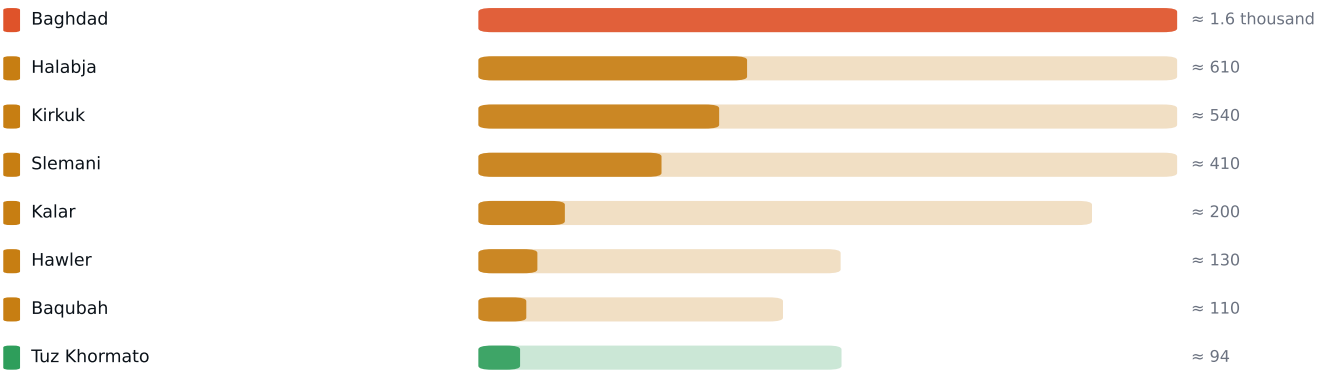
Share of each type's own buildings heavily damaged, worst first

Rubble stone, fieldstone	class A	**1.2%**	≈ 2.6 thousand buildings
Adobe, earth brick	class A	**0.8%**	≈ 2.0 thousand buildings
Manufactured stone units, timber floors	class B	**0.2%**	≈ 3.1 thousand buildings
Simple stone	class B	**0.2%**	≈ 2.0 thousand buildings
RC flat slab and others	class B	**0.2%**	≈ 12 buildings
Manufactured stone with RC floors	class C	**0.1%**	≈ 2.4 thousand buildings
RC frame — no / low code (ERD-L)	class C	**0.1%**	≈ 380 buildings

Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Slemani	668 thousand	4.6 thousand	1.5 thousand to 25 thousand	0.69%
Diyala	605 thousand	3.3 thousand	820 to 21 thousand	0.54%
Baghdad	2.1 million	1.7 thousand	480 to 11 thousand	0.08%
Kirkuk	568 thousand	1.2 thousand	270 to 9.5 thousand	0.20%
Salah Al-Din	592 thousand	830	220 to 6.7 thousand	0.14%
Hawler	659 thousand	300	110 to 2.0 thousand	0.05%
Wassit	376 thousand	280	82 to 2.0 thousand	0.07%
Babil	534 thousand	260	78 to 2.0 thousand	0.05%
Ninewa	731 thousand	260	82 to 1.6 thousand	0.03%
Al-Anbar	359 thousand	160	51 to 1.1 thousand	0.05%
Kerbala	314 thousand	100	36 to 660	0.03%
Duhok	234 thousand	28	9 to 140	0.01%
Maysan	17 thousand	8	2 to 69	0.05%
Al-Najaf	19 thousand	5	1 to 37	0.03%

How this report was made

Hazard engine Bumelerze SHAKEmap 26.2.0, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)

Observations 0 seismic stations, 1131 felt-report cells (felt channel: dyfi)

Damage scale IMS-25

Data sources

Built from USGS, EMSC and GEOFON earthquake parameters; building footprints from GFZ OpenBuildingMap (c) OpenStreetMap contributors, ODbL, and Microsoft, CDLA-Permissive-2.0; JRC Global Human Settlement Layer and WorldPop, CC BY 4.0; OCHA COD-AB Iraq, CC BY-IGO; GEM Global Exposure Model; IMS-25 vulnerability table. Full records, with licences and checksums: github.com/Peshawa-LH/bumelerze-engine

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Hasan P. L. (2026) Bumelerze SHAKEmap and damage report, event bml20170001, version 2, generated 19 Oct 2026, 13:26 UTC.